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BIOSTATISTICS REVIEW

FOR USMLE STEP 1, 2CK, 3

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BIOSTATISTICS QUESTIONS

1. 62-year-old male comes to the office for a check-up. He has no significant past medical history and currently feels fine. Last year his wife was diagnosed with adenocarcinoma of the colon. He is interested in the screening programs for colon cancer. Which of the following features of a test makes it valuable as a screening tool:

   a) high specificity
   b) high positive predictive value
   c) high sensitivity
   d) high accuracy
   e) high precision

   The correct answer is c.

   The idea behind screening is to identify early cases of a disease. In order for a test to be a valuable screening test it needs to identify almost all individuals affected by a condition. As such, it is important that screening tests have the lowest possible false negative rate. The false positive rate is of lesser concern as individuals identified by screening tests can be further assessed by tests with higher specificity to confirm the diagnosis. Positive and negative predicted value tell us what proportion of positive and negative tests are true positives/negatives. Accurate tests are the ones that produce results that are close to the real value, while precise tests produce results that are close to each other.

2. A large medical center has developed a biomarker that can be used for ALS screening. They decide to offer the test to all patients aged over 50. Subsequently, they determine that following the introduction of the test, survival of ALS patients has increased by approximately two months. Which of the following is a possible explanation of the finding:

   a) selection bias
   b) measurement bias
   c) length-time bias
   d) lead-time bias
   e) confounding

   The correct answer is d.

   Lead-time bias refers to apparent survival increase due to earlier disease detection without any change in disease course. ALS is an incurable disease with very limited treatment options. Early treatment would be unlikely to alter the disease course. Selection bias refers to nonrandom sampling of study participants. Measurement bias would be present if there was an issue with the way a variable was measured (eg, faulty equipment). Length-time bias refers to the tendency of screening tests to detect diseases with longer latency periods (eg, the window in which a slowly-progressive lymphoma that may have a latency period of a many years can be diagnosed is far larger than the one for rapidly-progressive lymphomas). Confounders are external variables that may affect the outcome. For example, if heroin users were more likely to be alcoholics, it could be determined that alcoholic liver disease is more common among heroin users and lead to the conclusion that heroin causes alcoholic liver disease.

3. 62-year-old female with history of hypertension presents to her primary care physician's office to discuss her blood pressure management. For the past week, she has kept a blood pressure diary by measuring it at the same time each day. Her systolic blood pressure measurements (in mmHg) were: 145, 129, 140, 129, 129, 132, 145. Which of the following values represents her median systolic blood pressure over the past week:

a) 145
b) 132
c) 140
d) 136
e) 129

The correct answer is b.

The median measurement is the one that has an equal number of measurements greater and lesser than it. In this case it is 132 as 3 of the measurements were 129, which is lower than 132 and 3 measurements were 136, 140 and 145 all of which are larger than 132. In case there is an even number of measurements, the median is the average of the two middle values. The mode is the most frequently encountered value, which in this case is 129. The mean value is the sum of all the measured values divided by the number of measurements: (145+129+140+129+129+132+145)/7=136.

4. A case-control study examining the relationship between alcohol use and pancreatitis development generated the following data:

Alcohol users who developed pancreatitis: 60; alcohol users who did not develop pancreatitis 30, alcohol non-users who developed pancreatitis: 40; alcohol non-users who did not develop pancreatitis: 70. Which of the following represents the way odds ratio is calculated:

a) (60*40)/(70*30)

b) (60*30)/(70*40)
c) (60*70)/(40*30)
d) (60*70)/(60*30*40*70)
e) cannot be determined

The correct answer is c.

The odds ratio represents the odds that a person affected by a condition was exposed to a risk factor (odds 1) versus the odds that a person unaffected by the same condition was exposed to the same risk factor (odds 2). It is calculated as odds 1/odds 2. Odds 1 = 60/40; odds 2 = 30/70. A simple calculation gives us the OR = (60*70)/(40*30).

5. An epidemiology graduate student has chosen to write her PhD thesis in epidemiology on the topic of disease prevalence and incidence. For the purposes of the study, she is required to pick a disease with an approximately equal prevalence and incidence. Which of the following diseases best satisfies the requirement:

a) Streptococcal pharyngitis
b) Aortic stenosis
c) HIV
d) Schizophrenia
e) Carpal tunnel syndrome

The correct answer is a.

Disease prevalence refers to the number of people affected by the condition, while disease incidence refers to the number of newly diagnosed cases. Therefore, in order for those two values to be close to each other, it is essential that the disease duration is short. Almost all patients recover from Streptococcal pharyngitis in the same year they develop the disease, making the prevalence and incidence very closely aligned. Schizophrenia, carpal tunnel syndrome, aortic stenosis and HIV are all chronic conditions and once patients are diagnosed they generally remain affected for life (unless, for example, carpal tunnel syndrome is treated surgically). While incidence of those conditions may remain stable, with each passing year, the prevalence of the disease increases as previously diagnosed patients remain affected and new cases are diagnosed.

6. 68-year-old female has been diagnosed with pneumonia caused by the newly described coronavirus. Having watched the news, she is extremely worried about dying from the disease

and asks the physician how likely she is to survive the disease. Which of the following terms is she most interested in:

a) number needed to treat
b) number needed to harm
c) attributable risk
d) case fatality rate
e) relative risk

The correct answer is d.

Case fatality rate is the percentage of patients who die from the disease among those who have been affected. Considering that the patient wants to know how likely she is to survive the condition, the most valuable piece of information is the percentage of patients affected who end up dying, which is the case fatality rate. Relative risk is the risk of developing a disease among patients exposed to a risk factor divided by the risk in the unexposed (eg, it tells us that smokers are a certain number of times more likely to develop lung cancer compared to non-smokers). Attributable risk is the difference in risk between the two groups. Number needed to treat and number needed to harm refer to the number of patients who need to receive an intervention in order for 1 patient to be helped or harmed.

7. The investigators are planning a decade-long research project on the association between chronic kidney disease (CKD) and long-term NSAID use. The participants have been divided into two groups based on whether they are on chronic NSAID therapy. They will be followed for CKD development over the next ten years. This study can best be described as:

a) Prospective cohort study
b) Retrospective cohort study
c) Case-control study
d) Cross-sectional study
e) Clinical trial

The correct answer is a.

The project described is a textbook example of a prospective cohort study, where participants are divided into two groups based on their exposure to the risk factor and then followed for a period of time for development of an effect. In this case, the ultimate goal will be to determine the relative risk of CKD development in patients taking NSAIDs. Retrospective cohort studies are similar but their starting point is in the past. Their major advantage over prospective studies is the significantly shorter timeframe required to complete the study. Clinical trials require an active intervention (eg, if NSAIDs were actively prescribed to protect from CKD, rather than given as a treatment for another condition). Neither cross-sectional nor case-control studies involve following participants for a period of time.

8. In a case control study, 200 patients with lung cancer were compared to 200 controls. They were asked about their smoking habits. The calculated odds ratio was 5.5; the 95% confidence interval was 3.2 to 7.8 for this association. Which of the following is the correct interpretation of these results:

a) there is a 95% chance that the odds ratio is 5.5 and 5% chance that it is between 3.2 and 7.8
b) at least 5% of smokers develop lung cancer
c) 5% of smokers account for 7.8% of newly diagnosed lung cancers
d) there is a 95% chance that the true value of the odds ratio is between 3.2 and 7.8
e) there is on average a 5.5% increased risk for lung cancer among smokers

The correct answer is d.

The confidence interval represents an interval where the true value most likely falls. There is a 95% chance that the true value falls within the 95% confidence interval. In order for results to be statistically significant, it is essential that it does not contain the value 1. In this case, the results are statistically significant and there is a 95% chance that the true odds ratio falls between 3.2 and 7.8.

9. Researchers are looking for new biomarkers that can aid in diagnosing congestive heart failure (CHF) in the areas with a shortage of cardiologists, where obtaining echocardiograms may be difficult. They have identified five different molecules and named them CHF1, CHF2, CHF3, CHF4, and CHF5. Their sensitivities and specificities were compared to findings on echocardiography, which is considered the gold standard. Which of the biomarkers would be the most useful to confirm the diagnosis of CHF:

a) CHF1 - sensitivity: 67%, specificity 72%
b) CHF2 - sensitivity 95%, specificity 56%
c) CHF3 - sensitivity 81%, specificity 81%
d) CHF4 - sensitivity 50%, specificity 89%
e) CHF5 - sensitivity 58%, specificity 93%

The correct answer is e.

The higher the specificity of the test, the better the test is for confirming the diagnosis. In contrast, the higher the sensitivity of the test, the better it is as a screening test. The goal of screening is to identify all patients with the condition, even at the expense of getting a number of false positive results. Confirmatory tests, on the other hand, are best when the rate of false-positives is as low as possible, meaning that the patients who test positive very likely have the disease.

10. 25-year-old male presents to the office with fever and sore throat for the past 3 days. His past medical history is insignificant. The physician performs an exam and concludes that streptococcal pharyngitis is the most likely explanation of the patient's symptoms. To confirm the diagnosis, he decides to use a novel test which has a positive predictive value of 87%. The patient tests positive. Which of the following best explains the meaning of this finding:

a) there is a 13% chance the patient has the disease
b) there is a 13% chance the patient does not have the disease
c) the test is 87% accurate in diagnosing the disease
d) the test is 87% specific for the disease
e) the test is 87% sensitive for the disease

The correct answer is b.

The positive predictive value of a test is a probability that a person who tests positive indeed has the condition. The test the physician in question used reportedly has a positive predictive value of 87%, meaning there is an 87% chance that the patient who tested positive has the disease. That, in turn, means that there is a 13% chance that he does not have the disease. Accuracy of the test refers to the tests ability to yield results that are close to the real value. Sensitivity of the test is the probability that the test is positive when the disease is present, while the specificity is the probability that the test is negative when the disease is absent.

11. A health department of a small island nation with a steady population and almost no migration is writing the annual report on type II diabetes in the community. There are 12,000 people living on the island, with 1,800 of them currently affected by the disease. 100 of them were diagnosed in the previous year. In the same year 80 patients with diabetes passed away. Which of the following is the prevalence of the disease in the population:

a) 100/1,800
b) 100/12,000
c) 1,800/12,000
d) 20/1,800

e) 100/10,200

The correct answer is c.

The prevalence of a disease in the population tells us the percentage of the population that is affected by a condition. It is calculating by dividing the number of existing cases with the total number of people in the population. In this population 1,800 out of 12,000 people have type II diabetes. Therefore, the prevalence of the disease is 1,800/12,000. The incidence of the disease looks at the number of new cases (in this case 100) and is calculated by dividing that number by the number of people at risk. The fact that 100 new cases of the disease were diagnosed and that only 80 patients passed away means that the prevalence of the disease is increasing even with a steady incidence, which is typical of many chronic diseases.

12. A medical student is writing his first research paper. He is interested in finding out whether there is an association between acute pancreatitis and paracetamol use. He will be doing a retrospective cohort study and has already identified the patients that will be enrolled in the study. Which of the following is the correct null hypothesis for his study:

a) There is a statistically significant association between acute pancreatitis and paracetamol use
b) Paracetamol use leads to acute pancreatitis
c) Acute pancreatitis leads to increased paracetamol use
d) There is no association between paracetamol use and acute pancreatitis
e) Paracetamol use protects from acute pancreatitis

The correct answer is d.

The null hypothesis is the hypothesis of no relationship. The alternative hypothesis is the hypothesis of some relationship. The null hypothesis is defined prior to the beginning of the study. The study then tests it and concludes with it being either accepted or rejected. In this case, the null hypothesis is that there is no association between paracetamol use and acute pancreatitis. The alternative hypothesis is that there is a statistically significant association between acute pancreatitis and paracetamol use. If the study finds that there is a statistically significant association between the two, the null hypothesis is rejected. If the study finds that there is no statistically significant association between the two, the null hypothesis is accepted.

13. 64-year-old male presents to the office for his annual diabetes check-up. His past medical history is significant for type II diabetes, hypertension and hypercholesterolemia. He currently feels fine. Over the past week, he has been regularly checking his blood pressure levels. His

early morning readings were as follows (in mg/dL): 105, 108, 103, 99, 111, 108, 110. Which of the following is the mode of the readings:

a) 99
b) 105
c) 106
d) 108
e) 111

The correct answer is d.

Mode refers to the most commonly encountered value. In this case only 108 shows up twice and, therefore, it is the mode. The average is approximately 106 (sum of the values divided by the number of measurements). The median is the value that falls in the middle of all the measurements, meaning that there is an equal number of measurements greater than, and lesser than the median.

14. A new drug for the treatment of Alzheimer's disease was developed. It was shown that it increases the average survival by approximately 2 years. If it became the new standard of care, which of the following changes in prevalence and incidence of Alzheimer's disease would be expected:

a) incidence will not change, prevalence will not change
b) incidence will increase, prevalence will increase
c) incidence will increase, prevalence will not change
d) incidence will not change, prevalence will increase
e) incidence will decrease, prevalence will not change

The correct answer is d.

Increased survival would lead to the increased prevalence of the disease since prevalence refers to the total number of people affected by a condition divided by the total population. As people affected by the disease live longer and new cases are being diagnosed, the total number of people with the condition increases. Incidence refers to the number of newly diagnosed cases and would, therefore, not be affected by a improved management.

15. A new biomarker for COPD, PD1 has been discovered in a large academic hospital. Upon further investigation, it is discovered that among all individuals in the population, the values of

PD1 are normally distributed, with the mean of 100 mg/dL and SD of 20 mg/dL. Patients with PD1 values less than 60 mg/dL are considered to have tested positive and are likely to be affected by COPD. What percentage of the population would test positive in that scenario:

a) <1%
b) 2.5%
c) 5%
d) 16%
e) 32%

The correct answer is b.

PD values of 60 mg/dL correspond to 2 SDs below the mean (100-20*2). In normal distribution, approximately 68% of measurements fall within 1 SD of the mean. Therefore, around 16% fall outside of 1 SD from the mean on each side of the curve (32% in total). 95% of measurements fall within 2 SDs of the mean, which means that approximately 2.5% fall under 2 SDs below the mean. 99.7% of measurements fall within 3 SDs.

16. Researchers are investigating the association between smoking and mantle cell lymphoma, a rare but deadly form of lymphoma. At the end of the study they determine that the relative risk of developing mantle cell lymphoma in smokers over a 10 year period is 1.12 with the 95% confidence interval: 0.98-1.18. Which of the following conclusions is correct:

a) no significant association was found
b) smoking increases the risk of mantle cell lymphoma development
c) smoking decreases the risk of mantle cell lymphoma development
d) a confounder is present
e) in some patients smoking increases, while in others it decreases the risk of mantle cell lymphoma development

The correct answer is a.

The 95% confidence interval is an interval within which 95% of the time the real value would fall. If the RR equals 1, no association between variables exists. In order for results to be considered statistically significant, the confidence interval must not contain the number 1 (as it implies no correlation). In this case, the 95% confidence interval is between 0.98 (small decrease in risk) and 1.18 (increased risk) - therefore, 1 falls within. A confounder is an external variable that is related to both the cause and effect and affects the results of the study (eg, if the smokers are more likely to also consume alcohol, it could be falsely concluded that smoking causes alcoholic fatty liver disease).

17. A group of researchers is looking into developing a treatment protocol for carcinoma of the gallbladder, a rare type of tumor that usually affects the elderly. They are primarily interested in which chemotherapy regimens have led to increased survival. The clinic they work at rarely sees more than 2 gallbladder cancer patients per year. Which of the following types of study would be most suitable under the circumstances:

    a) clinical trial
    b) meta-analysis
    c) case report
    d) cohort study
    e) biased study

    The correct answer is b.

    Meta-analysis is a study done by pooling summary data from multiple different studies (which have to satisfy the inclusion/exclusion criteria of the study) and doing a statistical analysis of them in order to get more precise results. They are mainly useful for rare conditions where it would be impossible to find a sufficient number of patients for other forms of study. (Eg, imagine there are 10 cases per center. If we pool data from 20 different centers, the total number of cases becomes 200, rather than 10). Case reports only report on a single case. Cohort studies refer to studies done by comparing the incidence of an effect between a group of patients exposed and a group unexposed to a risk factor. Biased studies are those where due to an error in the way data was obtained or interpreted, the results of the study do not correspond to the actual values.

18. A study was done to compare the effectiveness of the new diagnostic test for asthma to the current gold standard. 100 patients with confirmed asthma undergo the new test and 56 of them test positive. The same test is done on 100 patients who are confirmed not to have asthma and 12 of them receive a positive result. Which of the following represents the positive predictive value of the test:

    a) 56%
    b) 67%
    c) 82%
    d) 88%
    e) cannot be determined

The correct answer is c.

Positive predictive value is the probability that the person who tests positive has the condition in question. In this case, there were 56 true positives (people with the disease who tested positive) and 12 false positives (people who do not have the condition who tested positive). The positive predictive value is calculated as TP/(TP+FP) = 56/(56+12) = 0.82, or 82%. The negative predictive value of the test is the probability that the person who tests negative actually does not have the disease. In this case, there were 88 true negatives (people without asthma who tested negative) and 44 false positives (people who do not have asthma but tested positive). The negative predictive value is calculated as TN/(TN+FN) = 88/(44+88) = 0.67, or 67%. Sensitivity of the test is the probability that when the disease is present, the patient will test positive. It is calculated as TP/(TP+FN) = 56/100 = 0.56, or 56%. Specificity is the probability that when the disease is absent, the patient will test negative. It is calculated as TN/(TN+FP) = 88/100 = 0.88, or 88%.

19. Which of the following is the negative predictive value of the test:

a) 56%
b) 67%
c) 82%
d) 88%
e) cannot be determined

The correct answer is b.

The negative predictive value of the test is the probability that the person who tests negative actually does not have the disease. In this case, there were 88 true negatives (people without asthma who tested negative) and 44 false positives (people who do not have asthma but tested positive). The negative predictive value is calculated as TN/(TN+FN) = 88/(44+88) = 0.67, or 67%. Positive predictive value is the probability that the person who tests positive has the condition in question. In this case, there were 56 true positives (people with the disease who tested positive) and 12 false positives (people who do not have the condition who tested positive). The positive predictive value is calculated as TP/(TP+FP) = 56/(56+12) = 0.82, or 82%. Sensitivity of the test is the probability that when the disease is present, the patient will test positive. It is calculated as TP/(TP+FN) = 56/100 = 0.56, or 56%. Specificity is the probability that when the disease is absent, the patient will test negative. It is calculated as TN/(TN+FP) = 88/100 = 0.88, or 88%.

20. Which of the following is the sensitivity of the test:

a) 56%
b) 67%
c) 82%
d) 88%
e) cannot be determined

The correct answer is a.

Sensitivity of the test is the probability that when the disease is present, the patient will test positive. It is calculated as TP/(TP+FN) = 56/100 = 0.56, or 56%. Specificity is the probability that when the disease is absent, the patient will test negative. It is calculated as TN/(TN+FP) = 88/100 = 0.88, or 88%. Positive predictive value is the probability that the person who tests positive has the condition in question. In this case, there were 56 true positives (people with the disease who tested positive) and 12 false positives (people who do not have the condition who tested positive). The positive predictive value is calculated as TP/(TP+FP) = 56/(56+12) = 0.82, or 82%. The negative predictive value of the test is the probability that the person who tests negative actually does not have the disease. In this case, there were 88 true negatives (people without asthma who tested negative) and 44 false positives (people who do not have asthma but tested positive). The negative predictive value is calculated as TN/(TN+FN) = 88/(44+88) = 0.67, or 67%.

21. A study is done to assess the relationship between analgesic use and congestive heart failure (CHF) development. 1000 cases of CHF and 1000 controls of similar age, gender and social status are identified. They are asked to fill a survey detailing their history of analgesic use. The researchers conclude that CHF development is closely related with increased analgesic use, with an odds ratio approaching 3. Which of the following needs to be addressed before concluding that such an association exists:

a) selection bias
b) recall bias
c) observer bias
d) lead-time bias
e) measurement bias

The correct answer is b.

The study is done by asking patients to fill a survey. Such studies are prone to what is known as recall bias. Patients affected by a condition are often much more likely to remember specific instances of exposure to a risk factor and overreport it compared to controls. Selection bias would imply improper selection of study participants. The fact that controls of similar age, gender and social status were selected makes such bias unlikely. Confounders are external variables related to both the risk factor and the effect, which significantly change the findings in the study. Measurement bias happens due to faulty equipment or inappropriate data collection methods. If the researcher's conclusions were altered by their own beliefs (eg, a researcher who strongly believed that an association existed would be more likely to document findings that suggest an association), that would be considered observer bias.

22. A new study was developed to confirm the diagnosis of Hodgkin's lymphoma. Lymph node biopsy is considered the gold standard for diagnosing Hodgin's lymphoma. 1000 patients were tested by the new method before the biopsy was obtained. The following results were obtained: 432 patients tested positive on both the new study and the biopsy. 68 patients tested negative on the new test but were found to be affected after the biopsy was performed. 382 patients tested negative both times. 118 patients tested positive on the new test but their biopsy results were negative. What was the number of false positives in the study:

a) 432
b) 68
c) 382
d) 118
e) 500

The correct answer is d.

The number of false positives is the number of people who tested positive on the new test but were later determined not to have the disease. In this case, the number is 118. 382 is the number of true negatives - people who tested negative and were later confirmed to be negative. There were 432 true positives - people who tested positive and were confirmed to be affected. 68 was the number of false negatives - people who tested negative but were later found to have the disease. 500 was the total number of patients affected and the total number of patients who were not affected in this study.

23. What is the sensitivity of the previously mentioned test:

a) 86%
b) 76%
c) 79%
d) 85%
e) 100%

The correct answer is a.

The sensitivity of the test is calculated as TP/(TP+FN), in this case 86%. The specificity is calculated as TN/(TN+FP), in this case 76%. The positive predictive value is calculated as TP/(TP+FP), in this case 79%. The negative predictive value is calculated as TN/(TN+FN), in this case 85%

24. Which of the following conclusions can be reached about the new test:

a) It is as specific as the biopsy.
b) Most people who test positive do not have the disease.
c) It identifies the majority of people with Hodgkin's lymphoma.
d) 85% of people test negative.
e) It should replace the lymph node biopsy as the test of choice.

The correct answer is c.

The test has the sensitivity of 86%, which means that it is able to identify the majority of people who have the condition. With the specificity of the test at 76% it is not as specific as the biopsy. PPV of 79% means that the majority of people who test positive do have the disease. The numbers that we calculated do not tell us anything about whether the test should be used to replace the biopsy and in which circumstances. Those decisions are more complex and involve clinical, rather than purely statistical reasoning. The test does not tell us that 85% of people test negative. If the entire population were tested, the number of people who test negative would likely be far higher than 85% as relatively few people have Hodgkin's lymphoma.

25. A medical student is researching risk factors for bladder cancer. While reading the currently available literature he notices an unexpected trend - people who sit more and watch more TV seemed to be at an increased risk of bladder cancer. Which of the following is the best explanation of such findings:

a) inactivity leads to urinary stasis and consequently carcinoma
b) observer bias
c) measurement bias
d) people who watch more TV are more likely to report symptoms
e) smoking is a possible confounder

The correct answer is e.

Smoking is a known risk factor for bladder cancer. Confounders are variables related to both the risk factors and the disease that is being studied, which change the real association between the two. In this case, if people who sit more and watch more TV are also more likely to smoke, it could give the wrong impression that watching TV is associated with bladder cancer. Observer bias would be the explanation if the student's strong belief in the existence of an association was affecting the way he was analyzing data. There is no evidence of that. Measurement bias refers to faulty measurements leading to incorrect conclusions (eg, due to faulty equipment).

26. A 75-year-old male is being considered for the operative management of severe knee osteoarthritis. During the appointment, he expresses serious concern about the possibility of developing severe bleeding during the procedure. The available data suggests that 20% of patients develop significant bleeding during the procedure. Of those who do develop significant bleeding, 5% are considered serious. Which of the following best represents the patient's risk of developing severe bleeding:

a) 1/1,000
b) 2/1,000
c) 5/1,000
d) 1/100
e) 5/100

The correct answer is d.

20% of patients develop significant bleeding. Of them, 5% develop serious bleeding. 5% of 20% is 1%. Alternatively, if we assume that 1,000 patients have undergone the procedure, we can conclude that 200 of them (20%) developed significant bleeding. 5% of those patients developed severe bleeding. 5% of 200 patients is 10 patients. Overall, 10 out of 1000 patients (or 1/100) developed severe bleeding.

27. 63-year-old man presents to the office following the incidental discovery of an elevated PSA level on routine blood tests. The physician explains to him the need to obtain further testing, including a prostate biopsy. Which of the following features of the prostate biopsy makes it an ideal test to definitely confirm the diagnosis:

a) it is painless
b) it is highly sensitive

c) it is highly specific
d) complications from the procedure are rare
e) it is highly precise

The correct answer is c.

Specificity of the test refers to the probability that the test is negative when the disease is absent. Highly specific tests, therefore, have low false-positive rates. People who test positive can be considered affected by the condition. On the other hand, highly sensitive tests are used for screening as their main feature is the high probability that when the disease is present, the test will be positive (even at the expense of some false positive results). Highly precise tests produce results that are similar to each other. While clinical features of a test are taken into consideration when deciding to order it, they do not directly tell us how good a test is for either screening or disease confirmation. Furthermore, as far as diagnostic tests go, biopsies are neither among the least painful nor the least likely to cause complications.

28. A large clinical trial to compare the efficacy of a novel antihypertensive to that of ACE inhibitors is performed. Which of the following corresponds to the first phase of testing a new drug:

a) post-marketing surveillance
b) patients randomly assigned to the new drug or lisinopril
c) treating hypertension with the new drug on a small number of patients
d) small number of healthy volunteers given the new drug
e) animal studies

The correct answer is d.

Phase I of clinical trials refers to the phase where small numbers of healthy volunteers or patients are given the target drug. In the second phase, a somewhat larger number of patients is given the drug. In phase III researchers actively compare the target drug to placebo or the currently accepted gold standard treatment. Phase IV refers to surveillance after the treatment is approved in order to detect rare or long-term adverse effects which can lead to the drug being withdrawn from the market.

29. A 55-year-old female presents to the clinic for her regular mammography screening test. She feels fine and has no current complaints. Her family history is non-contributory. The test result comes back negative. The physician explains the meaning of the negative test and the likelihood

that she is indeed healthy. Which of the following features of a test depend on the prevalence of the disease in the population:

a) sensitivity and specificity
b) positive and negative predictive values
c) sensitivity and positive predictive value
d) specificity and accuracy
e) precision and negative predictive value

The correct answer is b.

The positive and negative predictive values of a test refer to the likelihood that people who test positive or negative are indeed positive or negative. Sensitivity and specificity tell us what the probability is that people who are positive or negative for the disease test positive or negative, respectively. Alongside accuracy and precision, they are inherent features of each individual test. PPV and NPV, on the other hand, depend on the prevalence of the disease in the population. Compare hypertension and silicosis for example. Considering how significantly more common hypertension is compared to silicosis, regardless of how people test for either condition, the likelihood that they are affected by hypertension is far higher. Therefore, the likelihood that those who test positive are affected is also higher.

30. A group of researchers is assessing the risk of developing myocardial infarction in a group of high-risk patients after switching to a DASH diet. The annual risk of developing a myocardial infarction in the group was 10% before the dietary change. In the following year, among those who successfully changed their diet, 5% developed myocardial infarctions. Which of the following represents absolute risk reduction of DASH diet:
a) 5%
b) 10%
c) 25%
d) 50%
e) cannot be determined

The correct answer is a.

Absolute risk reduction represents the difference in risk. In this case it is 10%-5%=5%. Relative risk reduction, on the other hand, refers to the proportion. In this case, it would be 50% (the risk was halved).

31. Which of the following represents the relative risk reduction in the same scenario:

    a) 5%
    b) 10%
    c) 25%
    d) 50%
    e) cannot be determined

    The correct answer is d.

    Relative risk reduction refers to what proportion of the risk has disappeared following the intervention. In this case, the risk was halved (from 10% to 5%) so the relative risk reduction is 50%. Absolute risk reduction represents the difference in risk. In this case it is 10%-5%=5%.

32. The researchers are considering using the ferritin value as the screening test for gastrointestinal malignancies. At the moment they are using the cut-off value of 20 ng/mL and referring patients with values lower than that for further testing. After a year, they changed the cut-off value to 50 ng/mL. Which of the following best describes the change in sensitivity and specificity of the test following such change:

    a) sensitivity increased, specificity decreased
    b) sensitivity unchanged, specificity increased
    c) sensitivity decreased, specificity unchanged
    d) sensitivity unchanged, specificity unchanged
    e) sensitivity decreased, specificity increased

    The correct answer is a.

    If the cut-off value is changed from 20 to 50 mg/dL, more people are going to test positive (as everyone with ferritin values between 20 and 50, who would have been considered negative a year ago, would be considered positive now). Therefore, the test will be more likely to identify the vast majority of cases but those who are identified will be less likely to have the disease. Therefore, the sensitivity of the test will increase but the specificity will decrease (the number of false negatives will decrease but the number of false positives will increase).

33. A large cohort study is performed to assess the efficacy of a novel statin in prevention of cardiovascular events. At the end of the study it is determined that 12% of patients treated with

older statins suffered a cardiovascular event compared to 8% treated with the new drug. Which of the following numbers corresponds to the absolute risk reduction in this study:

a) 67%
b) 33%
c) 4%
d) 8%
e) 12%

The correct answer is c.

Absolute risk reduction is the difference in risk. In this case the difference is 4% (12%-8%).

34. In the previously mentioned study, which of the following numbers corresponds to the relative risk reduction:

a) 67%
b) 33%
c) 4%
d) 8%
e) 12%

The correct answer is b.

Relative risk reduction is the proportion of risk reduction. In this case the risk on previous treatment was 12%, compared to 8% on the new treatment. That is a 33% risk reduction.

35. A medical student is working on his first research project. He has decided to analyze a disease that frequently affects his patients. For the purposes of the study, he needs to select a disease that has similar incidence and prevalence in the population. Which of the following conditions best satisfies the requirement:

a) atrial fibrillation
b) HIV
c) type II diabetes mellitus
d) depression
e) UTI

The correct answer is e.

Prevalence refers to the number of people currently affected by the condition, while incidence looks at the ratio of the number of new cases to the number of people at risk. Chronic conditions generally have prevalences much higher than incidences because each case is counted in the incidence statistic only once but remains counted in the prevalence statistic for years. Of the conditions in question, only UTI is an acute condition. Since almost all patients with UTI recover relatively quickly, they are calculated only once for both prevalence and incidence.

36. A 65-year-old female with past medical history significant for obesity, hypertension, hypercholesterolemia and type II diabetes mellitus is taking part in a clinical study assessing the effectiveness of a new weight loss drug compared to placebo. Patients are randomly assigned to either group and their weight is measured every day for three months. At the end of the study period, the results between the groups are compared. Which of the following is the correct null hypothesis for the study:

a) the new drug is more effective than placebo
b) placebo is more effective than the new drug
c) there is no significant difference in effectiveness between the new drug and placebo
d) the new drug causes weight gain
e) the new drug is associated with significant weight change in either direction

The correct answer is c.

The null hypothesis states that no difference exists. The alternative hypothesis states that a difference does exist. At the end of the study the null hypothesis is either accepted or rejected based on where the confidence interval falls.

37. In the previously mentioned study, which of the following would be a type II error:

a) wrongly concluding that the pill is associated with weight loss
b) wrongly concluding that the pill is not associated with weight loss
c) wrongly concluding the clinical significance of the findings
d) wrongly interpreting the results of measurements
e) failing to address possible biases in the study

The correct answer is b.

Type II errors refer to failing to reject a wrong null hypothesis. In this case, the null hypothesis states that there is no difference in weight loss between treatment and placebo. If the null hypothesis is false it means that there is a difference between the treatment and placebo. Failing to reject it would mean concluding that no difference exists. Type I errors refer to rejecting a correct null hypothesis (eg, concluding that there is a difference between the two groups when in fact there is none).

38. Which of the following could be done to decrease the probability of making a type II error:

a) rounding up the measurement results to the nearest 10
b) excluding a number of participants from the study
c) changing the null hypothesis
d) increasing the sample size
e) dividing the treatment group in two

The correct answer is d.

Beta is the probability of making a type II error and it is related to the power of the study (1-beta). Type II error refers to failing to reject the null hypothesis when it is, in fact, false (i.e., false-negative error; meaning, there is a difference but we said there’s not). Increasing the sample size, improving the precision of the measurement and increasing the expected effect size would increase the power of the study (and decrease beta).

39. The patient has also been checking her blood sugar levels daily over the same time period. Her blood sugar levels were mostly around 120 mg/dL. However, at one point, she was admitted to the hospital severely dehydrated and disoriented. At the time of the admission blood glucose was 526 mg/dL. Which of the following best describes the effect this measurement is having on the mean, median and mode of her blood glucose levels:

a) mean < median < mode
b) median < mean < mode
c) mean < mode < median
d) mode < mean < median
e) mode < median < mean

The correct answer is e.

A value that differs significantly from the usual measurements affects the measures of central tendency to a different degree. Mean is affected the most because in order for it to be calculated, we need the sum of all the values that were measured. Median is affected significantly less because it only adds one value to the high end and the median is the value that falls in the middle of all measurements. The mode is affected the least as it is the most frequently measured value. In this case 526 mg/dL is not the most frequently measured value so the mode does not change.

40. A public health department is working on developing a program to decrease both the incidence and the prevalence of lung cancer. Which of the following interventions would best achieve that goal:

a) development of new chemotherapeutics
b) improving specialist referral times
c) introduction of a smoking cessation program
d) introduction of a screening program
e) increasing the number of surgeons

The correct answer is c.

If a risk factor (such as smoking) is eliminated it is going to lead to a decreased incidence of the disease. The consequence of that is that without any change in the way lung cancer is treated, the prevalence of the disease will also go down. Developing new chemotherapeutics, improving specialist referral times, introducing screening programs and increasing the number of surgeons could all improve patient outcomes but would not alter the incidence of the disease. If those measures led to improved survival, the prevalence of the disease could increase.

41. A study is being done to assess the strength of association between smoking and coronary disease development. At the beginning of the study, the patients are divided into two groups based on whether they smoke cigarettes on daily basis. They were then followed for 10 years for the development of any form of coronary heart disease. At the end of the study period, there does not appear to be a significant difference in coronary disease incidence between the two groups. Which of the following most likely explains the finding:

a) smoking is not a risk factor for coronary disease
b) Berkson bias
c) observer bias

d) recall bias
e) Hawthorne effect

The correct answer is e.

Hawthorn effect refers to a behavioral change in participants who know they are being followed. In this case it is possible that patients who were followed stopped smoking or decreased the amount they smoked. Smoking is a known risk factor for coronary disease development. Observer bias could explain the finding if the researchers strongly believed that there was no association between the two and if such belief affected the way they interpreted the results. In this case, the likelihood that the participants simply changed their behavior in responds to being followed is higher. Recall bias is a term used to describe the phenomenon where people who develop a condition tend to overreport exposure to a risk factor that is being studied compared to controls. This study does not depend on the participants' recall of risk factor exposure. Berkson bias refers to the tendency of hospitalized patients to be less healthy and have different exposures when compared to the general population. The participants in this study were not necessarily hospitalized patients.

42. A group of researchers is is conducting a study to compare a new anti-diabetic drug with metformin. A total of 500 patients were enrolled in a study, 300 received the new drug, while 200 received metformin. Both groups were followed for 3 months. At the end of the 3-month period, the average fasting blood glucose level in the group treated with metformin was 135.5, while the average blood glucose level in the group treated with the new drug was 133.9 ($p=0.03$). Which of the following represents the best interpretation of those findings:

a) the new drug resulted in clinically insignificant but statistically significant effect
b) the new drug resulted in clinically significant but statistically insignificant effect
c) the new drug resulted in clinically significant and statistically significant effect
d) the new drug resulted in clinically insignificant and statistically insignificant effect
e) cannot be determined from the data given

The correct answer is a.

The fact that the p value is <0.05 suggests that the new drug has resulted in a statistically significant decrease in blood glucose levels. However, the actual clinical value of decreasing the fasting blood glucose from 135.5 to 133.9 is unlikely to be significant.

43. A large study is done to assess the average blood glucose values between patients with different chronic conditions. Patients are divided into groups based on what their underlying co-morbidity is and their blood glucose measurements are taken. At the end of the study, the researchers want to compare the mean blood glucose levels between the groups. Which of the following statistical tests would be the most appropriate:

a) t-test
b) ANOVA
c) Chi-square
d) Fisher's exact test
e) meta-analysis

The correct answer is b.

ANOVA test is used to compare the means between the groups. t-test is used to compare the means of two groups. Chi-square and Fischer's exact test are used to check differences between categorical outcomes. Meta-analysis is a type of study where data is gathered from a large number of previous studies and statistically analyzed in order to get a more precise result.

44. If the researchers from the previous question wanted to compare blood glucose level between healthy subjects and those with any chronic condition and only used the groups: healthy and not healthy, which of the following tests would be the most appropriate:

a) t-test
b) ANOVA
c) Chi-square
d) Fisher's exact test
e) meta-analysis

The correct answer is a.

t-test is used to compare the means of two groups. ANOVA test is used to compare the means between the groups. Chi-square and Fischer's exact test are used to check differences between categorical outcomes. Meta-analysis is a type of study where data is gathered from a large number of previous studies and statistically analyzed in order to get a more precise result.

45. A study was done to determine the incidence and outcomes of myocardial infarction in a large urban area. The annual incidence of myocardial infarction was found to be 600 per 100,000 people with the case fatality rate of 15%. Which of the following represents the annual mortality from myocardial infarction in this population:

a) 150/100,000
b) 1,500/100,000
c) 900/100,000
d) 60/100,000
e) 90/100,000

The correct answer is e.

The case fatality rate is the proportion of people affected by a condition who die from it. In this population 15% of those who suffer myocardial infarctions die from them. The annual incidence of the disease is 600/100,000. Therefore, the mortality from the condition is 15% of 600/100,000 or 0.15*600/100,000 = 90/100,000.

46. A resident is pre-rounding the 7 patients under his watch. As part of the check-up he measures their heart rates. He measured the following values (per minute): 92, 100, 83, 61, 86, 83, 102. Which of the following is the mean heart rate that he measured:

a) 83
b) 86
c) 87
d) 61
e) 102

The correct answer is c.

The mean is calculated by dividing the sum of all the measurements by the number of measurements. In this case the exact result is 86.7, which can be rounded to 87. 61 and 102 are the lowest and highest value, respectively. 83 is the mode, or the value that appears most frequently. 86 is the median or the value that has an equal number of measurements falling above and below it.

47. What is the median heart rate that the resident measured:

a) 83
b) 86
c) 87
d) 61
e) 102

The correct answer is b.

86 is the median or the value that has an equal number of measurements falling above and below it. The mean is calculated by dividing the sum of all the measurements by the number of measurements. In this case the exact result is 86.7, which can be rounded to 87. 61 and 102 are the lowest and highest value, respectively. 83 is the mode, or the value that appears most frequently.

48. A new ACE inhibitor was developed specifically for slowing the progression of advanced diabetic nephropathy over 5 years. In a randomized clinical trial, the rates of ESRD development in the group treated with the new drug were 0.03. The odds of ESRD development in the group treated with ramipril were 0.12. Which of the following represents the relative risk reduction:

a) 3%
b) 9%
c) 12%
d) 25%
e) 75%

The correct answer is e.

The new drug has decreased the incidence of ESRD from 0.12 to 0.03 (12% to 3%). That represents a 75% drop in the number of new cases of ESRD. Relative risk reduction represents the proportion of risk reduction, while absolute risk reduction represents the difference in risk.

49. A large cohort study was done to assess the relationship between smoking and pancreatic cancer development. Patients were divided into smokers and non-smokers and followed for 10 years for pancreatic cancer development. At the end of the study period the relative risk was calculated. Which of the following confidence intervals would most clearly suggest that smoking significantly increases the risk of pancreatic cancer development:

a) 0.3-0.8
b) 0.5-1.1
c) 0.1-3.6
d) 0.9-2.5
e) 1.5-1.8

The correct answer is e.

In order for the results of a study to be statistically significant, the confidence interval of the relative risk must not contain the number 1. RR of 1 implies that there is an equal risk of developing the condition between the two groups. As such, for as long as 1 falls within the confidence interval, the results cannot be considered statistically significant, regardless of what the high and low end of the interval are. RR<1 implies that the factor that was studied was protective, while the RR>1 implies that it is a risk factor. In this case, only the 1.5-1.8 confidence interval clearly shows a statistically significant increase in risk of pancreatic cancer development.

50. Which of the following confidence intervals would indicate that smoking is protective against pancreatic cancer:

a) 0.3-0.8
b) 0.5-1.1
c) 0.1-3.6
d) 0.9-2.5
e) 1.5-1.8

The correct answer is a.

In order for the results of a study to be statistically significant, the confidence interval of the relative risk must not contain the number 1. RR of 1 implies that there is an equal risk of developing the condition between the two groups. As such, for as long as 1 falls within the confidence interval, the results cannot be considered statistically significant, regardless of what the high and low end of the interval are. RR<1 implies that the factor that was studied was protective, while the RR>1 implies that it is a risk factor. In this case, only the 0.3-0.8 confidence interval clearly shows a statistically significant decrease in risk of pancreatic cancer development.

51. A clinical trial is done to assess the effectiveness of a new statin drug. Patients are randomly divided into two groups. One of the groups is treated with the new drug for 3 months, while the other is treated with atorvastatin. Neither the participants nor the researchers know who is getting the new drug. Such approach best prevents which of the following:

a) confounding
b) selection bias
c) length-time bias
d) recall bias bias
e) observer bias

The correct answer is e.

Blinding in clinical trials prevents observer bias. If the researchers are aware of which patients are receiving the tested drug, there is a chance this knowledge may influence the way they interpret the data. Recall bias is a feature of case-control studies and refers to the tendency of participants to overreport exposure when the investigated effect develops. Selection bias refers to the way participants were selected for the study. Blinding has no effect on it. Length-time bias refers to the tendency of screening tests to detect diseases with longer latency periods. Confounders are external factors related to both variables being studied which can lead to incorrect interpretation of results.

52. A large clinical institute has developed a new drug for the management of advanced melanoma. The early results show the average survival time increasing by two years. For the most part, serious side effects were not observed and no participants died from complications of the treatment. If such treatment was administered to everyone with advanced melanoma, which of the following best describes the effect this would have on melanoma incidence and prevalence:

a) incidence would increase, prevalence would not change
b) incidence would not change, prevalence would decrease
c) incidence would decrease, prevalence would increase
d) incidence would not change, prevalence would increase
e) incidence would not change, prevalence would not change

The correct answer is d.

Finding a new treatment that improves survival for a condition would increase the prevalence of the disease in the society. Prevalence tells us how many people are affected. If people were to mostly die within a year from the time they were diagnosed, the prevalence and incidence would remain almost the same. However, if people live longer, despite the constant number of new cases, the total number of cases (new + old) keeps increasing. The incidence, meanwhile, would not change as it refers to the number of new cases that are being diagnosed. The way the disease is treated does not affect the number of people who develop it. A way to change the incidence of the condition would be to change risk factor exposure (eg, incidence would decrease if people abstained from spending excessive amounts of time in the sun without appropriate protection).

53. A 60-year-old male presents to the physician's office after his brother was diagnosed with colorectal cancer. His family history is otherwise non-contributory and he is currently in good health. The physical exam reveals no abnormalities. The patient undergoes screening colonoscopy which comes back negative. The patient is still concerned about the possibility a tumor was missed on a colonoscopy. While helping the patient understand the results, which of the following features of the test should the physician refer to:

a) precision
b) sensitivity
c) specificity
d) positive predictive value
e) negative predictive value

The correct answer is e.

The negative predictive value is the probability that the person with a negative test result is actually negative for the disease. In this case the patient tested negative and his concern is the probability that the test gave him a false negative result. The higher the negative predictive value of the test, the more likely it is that the patient is indeed healthy. The positive predictive value tells us what the probability is that someone who tests positive has the condition. Sensitivity and specificity tell us what percentage of people with or without the condition test positive or negative, respectively. Precise tests are the ones that give results that are close to one another.

54. The new definition of hypercholesterolemia has changed and the cut off value was decreased by 10 mg/dL. Which of the following best describes the effect such change would have on incidence and prevalence of hypercholesterolemia:

a) incidence unchanged, prevalence unchanged
b) incidence increased, prevalence increased
c) incidence increased, prevalence decreased
d) incidence decreased, prevalence decreased

e) incidence decreased, prevalence decreased

The correct answer is b.

Changing the definition so that more people would be seen as positive would increase both prevalence and incidence of the disease in question - incidence increases as more new cases will be diagnosed (due to the lower cut-off value) and prevalence increases as it is a chronic condition.

55. A group of researchers has started developing a new drug for the treatment of Acute Myeloid Leukemia. The molecule has been developed in the lab. 10 healthy volunteers will be given the drug. Such a scenario corresponds to which of the following phases of clinical trials:

a) phase I
b) phase II
c) phase III
d) phase IV
e) phase V

The correct answer is a.

There are 4 commonly identified phases of clinical trials. In phase I of the clinical trial a small number of healthy volunteers is given a drug to assess its safety, toxicity profile, pharmacodynamics and pharmacokinetics. In phase II a moderate number of patients is treated with the drug to assess its efficacy, optimal dosing and common side effects. In phase III a large number of patients is randomly assigned to treatment and placebo (or previously used gold standard treatment) groups. This phase compares the outcomes of patients given placebo or current standard of care to those of patients treated with the new drug. Phase IV refers to post-marketing surveillance to detect rare or long-term adverse effects.

56. Which of the following best describes what is being assessed in the previous question:

a) long-term safety of the drug
b) rare side effects
c) pharmacokinetics
d) optimal dosing
e) comparison with current standard of care

The correct answer is c.

There are 4 commonly identified phases of clinical trials. In phase I of the clinical trial a small number of healthy volunteers is given a drug to assess its safety, toxicity profile, pharmacodynamics and pharmacokinetics. In phase II a moderate number of patients is treated with the drug to assess its efficacy, optimal dosing and common side effects. In phase III a large number of patients is randomly assigned to treatment and placebo (or previously used gold standard treatment) groups. This phase compares the outcomes of patients given placebo or current standard of care to those of patients treated with the new drug. Phase IV refers to post-marketing surveillance to detect rare or long-term adverse effects.

57. It has been determined that the drug is safe. Which of the following would be the most appropriate next step for the researchers to take:

a) release the drug to the market
b) compare its effect to that of placebo
c) compare its effect to that of the current standard of care
d) give it to 30 patients diagnosed with ALS
e) give it to another 10 healthy volunteers

The correct answer is d.

Following the successful completion of phase I of a clinical trial, the drug enters phase II. In phase II the drug is given to a number of patients with the disease of interest in order to assess efficacy, adverse effects and determine optimal dosing. Comparing the drug to placebo or the current standard of care would only occur after that. Releasing the drug to the market is the appropriate step once the safety and efficacy of the drug have been proven by comparing it to the current standard of care.

58. Following a large clinical trial, a drug has been released to the market. Over the next few years, it proves to induce remission in a significant number of patients. However, there seems to be an increasing number of case reports pointing to seizure development following drug commencement. Which of the following would be the most appropriate response to such a situation:

a) withdraw the drug from the market
b) change the recommended dosage
c) change the recommended target patient population

d) administer it only together with anti-seizure treatment
e) administer the drug in the ICU

The correct answer is a:

Post-marketing surveillance refers to following the drug for years after it has been released to the market in order to determine less common and more long-term side effects. In this case, it appears that seizures are an uncommon but serious side effect. The most appropriate next step would be to withdraw the drug from the market until further analysis is done. The effective dosage has been determined in earlier phases and decreasing the dosage would likely also affect the efficacy of the drug without necessarily eliminating the unwanted side effect. While it may be the case that the drug is more damaging to a subset of the population, changing the target population without first withdrawing the drug from the market for further testing would not be the appropriate action. Adding an anti-seizure drug would increase the risk of other side effects. Chemotherapy is not administered in the ICU.

59. A study is done to assess the effectiveness of a new pain medication in treatment of chronic pain. A total of 100 patients were randomly assigned to either the new drug or no additional treatment. The participants were recruited from a pain clinic. They are then asked to complete a survey 7 days later asking them about their pain levels. A statistically significant improvement in pain levels is seen with patients on the new drug. Which of the following is a limitation to the validity of this study:

a) low number of participants
b) recall bias
c) recruiting patients from the pain clinic only
d) lack of blinding
e) subjective nature of the study

The correct answer is d.

Blinding refers to the unawareness of study participants (and/or researchers) as to whether they are receiving the treatment or not. In this case, the patients were aware of whether they were receiving an additional drug or not. Therefore, it is possible that due to placebo effect, patients who were receiving the drug could report significantly lower pain levels compared to those receiving no additional treatment. Recall bias refers to the tendency of patients who develop a disease to overreport their exposures. Since pain levels are subjective, a survey is a good way to evaluate the discomfort patients are experiencing.

60. A researcher is working on a project to determine the strength of association between smoking and bladder cancer development. He has decided to do a retrospective cohort study and has identified the appropriate study participants. At the end of the study he calculates the relative risk and confidence interval. The relative risk is 2.2. Which of the following p values for the outcome would suggest that the result is significant:

a) 0.04
b) 0.08
c) 0.10
d) 0.50
e) 1.00

The correct answer is a.

If the p value is <0.05 for a study outcome, there is less than a 5% chance that the result was due to chance alone. For clinical studies, the p value of 0.05 is generally used. Of the answer choices, only 0.04 is lower than 0.05.

61. A case-control study has been conducted to assess the relationship between excessive alcohol use and liver cancer development. 100 patients with liver cancer and 100 appropriately matched controls were identified. Of the 100 patients, 40 were found to have used excessive amounts of alcohol. Of the 100 controls, 15 were found to be using excessive amounts of alcohol. Calculating which of the following would be the most appropriate first step:

a) relative risk
b) odds ratio
c) attributable risk
d) number needed to harm
e) incidence

The correct answer is b.

Odds ratio is typically calculated in case-control studies. It represents the ratio of the odds of exposure among cases and the odds of exposure among controls. Relative risk is calculated in cohort studies and represents the risk of developing the disease in the exposed and unexposed groups. Attributable risk is the difference in risk between the exposed and unexposed group. Number needed to harm represents the number of patients who need to be exposed to a risk factor for 1 patient to be harmed. Incidence is the ratio of the number of new cases and the total number of people at risk.

62. Which of the following represents the correct way of calculating the odds ratio:

a) 100/40
b) 40/15
c) 100/15
d) 40*85/60*15
e) (40/100)/(85/100)

The correct answer is d.

The odds of exposure among cases are calculated as (exposed affected)/(unexposed affected). The odds of exposure among controls are calculated as (exposed unaffected)/(unexposed unaffected). The odds ratio is the ratio of the two odds, which equals 40*85/60*15. The choice e represents the way relative risk is calculated. RR would be calculated in a cohort study.

63. Which of the following biases needs to be addressed in this study:

a) lead-time bias
b) length-time bias
c) selection bias
d) procedure bias
e) recall bias

The correct answer is e.

Recall bias is common in retrospective studies that depend on patients' recall. In this case, it is possible that people diagnosed with liver cancer, aware of their disorder, may overreport their alcohol use when compared to unaffected individuals. Lead-time bias refers to the appearance of increased survival with early diagnosis that is not due to improved patient outcomes but rather purely due to an earlier date of disease detection. Length-time bias refers to the tendency of screening studies to better identify diseases with longer latency periods. Selection bias occurs due to nonrandom sampling of study subjects. The controls in this case were appropriately matched. Procedure bias occurs when subjects are not treated equally. This study was not assessing the management at all.

64. A group of researchers is comparing the efficacy of conservative and operative management of acute appendicitis. 500 patients are randomly assigned to one of the two groups - 250 into each group. The patients are then followed for 2 years. Which of the following best describes this study design:

a) case-control study
b) cross-sectional study
c) randomized clinical trial
d) prospective cohort trial
e) retrospective cohort trial

The correct answer is c.

The study that was described in the question stem is the randomized controlled clinical trial. It involves randomly assigning patients to one of two groups of interventions (or treatment versus placebo) and comparing the outcomes in the two groups. Cross-sectional studies are done to determine the prevalence of the disease. Case-control studies are done to determine the odds ratio, which is calculated as the odds of exposure among the affected group divided by the odds of exposure among the unaffected group. Both prospective and retrospective cohort studies are done to determine the relative risk of developing the disease among the exposed versus the unexposed group. In this case the patients randomly received one of two interventions and were followed to assess the efficacy and complication rates.

65. A study is done to assess blood pressure differences between residents of different states. A number of participants from each state is selected from the study. The mean blood pressure of residents of each state is calculated and the means are then compared. Which of the following statistical tests would be the most appropriate for such analysis:

a) t-test
b) ANOVA
c) Fischer’s exact test
d) Chi-square
e) meta-analysis

The correct answer is b.

ANOVA test is done to compare the means of 3 or more groups. t-test compares the means of two groups. Chi-square test checks differences between percentages or proportions of categorical outcomes. Fischer’s exact test checks differences between two percentages or proportions of categorical outcomes. Meta-analysis is a type of study where data from multiple studies is pooled and statistically analyzed to improve the power, strength of evidence and generalizability of the findings.

66. A 54-year-old female comes to the office to discuss preventive care. She is told that mammography is recommended for all women her age. The patient wants to learn more about what makes mammography a good choice of a test for her. In general, which of the following traits best fits a good screening test:

a) low number of false positives
b) low number of false negatives
c) high precision
d) low number needed to treat
e) low attrition rate

The correct answer is b.

The idea behind screening is to identify almost all individuals affected by a condition. The number of false positive is of lesser concern in this case as the final diagnosis can be confirmed with more specific tests later on. Precision refers to how close to each other the results of the same test are. The number needed to treat describes the impact of an intervention or exposure in cohort studies.

67. A retrospective cohort study was done to assess the relationship between long-term ibuprofen use and stomach cancer development. The relative risk that was calculated was 1.05 with the confidence interval between 0.99 and 1.08. Which of the following represents a correct conclusion from the study:

a) it cannot be concluded that long-term ibuprofen use is associated with stomach cancer development
b) long-term ibuprofen use has a small but clear correlation with stomach cancer development
c) long-term ibuprofen use is likely to be protective against stomach cancer
d) in some patients long-term ibuprofen use may decrease, while in others it may increase the risk of stomach cancer
e) there is stronger evidence in favor of long-term ibuprofen use causing stomach cancer than not causing it

The correct answer is a.

The relative risk of 1 implies that there is no association between long-term ibuprofen use and stomach cancer development. In order for results of a clinical study to be considered significant, the value 1 must not fall within the confidence interval as that would mean that there is <5% chance of there being no association. The RR values above 1 indicate that the factor that is being studied is associated with an increased incidence of the condition being studied, while values below 1 indicate that the opposite. In order for either of those conclusions to be reached, the entirety of the confidence interval must fall on the same side of the value 1.

68. A clinical study was done to assess the response to a novel calcium channel blocker in patients with essential hypertension. The drug was compared to amlodipine and it was found that on average it reduced the blood pressure by 2.3 mmHg more than amlodipine (95% confidence interval: 0.9-2.6 mmHg). Despite the findings, the drug has not replaced amlodipine as the preferred treatment. Which of the following explains the likely rationale behind such decision:

a) the results are not statistically significant
b) the results are statistically but not clinically significant
c) the study has not addressed enough parameters to be relevant
d) the drug needs to undergo more phases of a clinical trial
e) the decision not to replace amlodipine was wrong

The correct answer is b.

It is important to understand the difference between clinical and statistical significance. In this case, the confidence interval does not contain the value 0. Therefore, the results have reached statistical significance and the new drug does reduce blood pressure more than amlodipine. However, the blood pressure decrease of 2.3 mmHg is unlikely to lead to clinically improved outcomes. This drug has already undergone the third phase of clinical trial (comparison with current standard of care). The next step would be the release the drug to the market.

69. A 75-year-old female smoker with significant family history of lung cancer (patient A) and a healthy 25-year-old female non-smoker with no family history of lung cancer (patient B) undergo CT scans of the chest. Which of the following best represents the correct comparison of using the test for diagnosing lung cancer in these two patients:

a) the sensitivity of the test is higher for patient A
b) the specificity of the test is higher for patient A
c) the positive predictive value is higher for patient A
d) all of the above is higher for patient A

e) none of the above is higher for patient A

The correct answer is c.

Sensitivity and specificity are integral features of a diagnostic test and they do not depend on any external variables. Sensitivity is the likelihood that a person affected by a condition tests positive, while specificity is the likelihood that a healthy person tests negative. Positive and negative predictive values, on the other hand, do depend on external variables, such as disease prevalence in a certain population. Prevalence of lung cancer is much higher among 75-year-old smokers with family history of lung cancer than it is among 25-year-old non-smokers with no such family history. Therefore, the positive predictive value of the test is higher in patient A.

70. A group of researchers is trying to assess the relationship between exposure to second-hand tobacco and abdominal aortic aneurysm development. Their ultimate goal is to determine the relative risk of abdominal aortic aneurysm development in people exposed to significant amounts of second-hand smoke compared to those unexposed. They want to present the results at the conference in six months. Which of the following types of study would be best suited to their needs:

a) cross-sectional study
b) case-control study
c) retrospective cohort study
d) prospective cohort study
e) clinical trial

The correct answer is c.

Cohort studies are used to determine the relative risk of developing a condition based on exposure to a risk factor. Considering the time limitation, a retrospective cohort study makes more sense as prospective cohort studies take a long time to complete. Clinical trials require an active intervention, which, in this case, would be unethical. Case-control studies determine the odds of exposure in affected versus unaffected individuals. Cross-sectional studies determine the frequency of the disease and the risk factor.

71. A study is done to assess the relationship between smoking and congestive heart failure (CHF) development. At the end of the study, it is determined that the risk of CHF development is 5% in non-smokers and 15% in smokers. Which of the following is the attributable risk of smoking to CHF development:

a) 5%
b) 10%
c) 15%
d) 20%
e) 300%

The correct answer is b.

Attributable risk is the difference in risk between the 2 groups that are being studied. In this case it is 15%-5%=10%.

72. Three similarly priced, inexpensive tests were developed to evaluate for breast cancer in women. The sensitivity (Se) and specificity (Sp) of each test are given below. Which of the following testing strategies is the most appropriate:

Test 1: Se 89% Sp 96%
Test 2: Se 60% Sp 99%
Test 3: Se 98% Sp 72%

a) Test 3 followed by test 2 if test 3 is positive
b) Test 2 followed by test 3 if test 2 is positive
c) Test 1 followed by test 2 if test 1 is positive
d) Test 1 only
e) Test 3 followed by test 1 if test 3 is positive

The correct answer is a.

The appropriate order of tests is from more sensitive to more specific tests. Sensitive tests have very low rates of false negatives, whereas specific tests have low rates of false positives. That is why sensitive tests are good for screening, while specific tests are used for confirmation. In this case, test 3 is the most sensitive and should, therefore, be done first. Test 2 is the most specific and is the best test to confirm the diagnosis. If only one test could be performed, then using test 1 could be a reasonable alternative.

73. A study is done to assess blood pressure control among patients treated with thiazide diuretics. At the end of the study it is discovered that the device being used was not functioning properly. Which of the following correctly identifies the resulting bias in the study:

a) procedure bias
b) confounding bias
c) observer bias
d) measurement bias
e) length-time bias

The correct answer is d.

Measurement bias occurs when faulty equipment leads to incorrect readings and therefore biased study results. Procedure bias occurs when participants are not treated equally. Confounders are external variables related to both factors that are being studied that may lead to incorrect conclusions about study results. Observer bias occurs when the researcher’s strong belief in the existence or non-existence of the association being study leads to incorrect interpretation of the study results. Length-time bias refers to the tendency of screening tests to better identify diseases with longer latency periods.

74. There are currently 4,000 people suffering from multiple sclerosis in a city with the population of 1 million people. 300 new cases are diagnosed each year and 40 deaths can be attributed to multiple sclerosis. In total, approximately 10,000 people die in that city every year. Which of the following represents the prevalence of multiple sclerosis:

a) 300/1,000,000
b) 4,000/1,000,000
c) 40/1,000,000
d) 300/996,000
e) 4,000/996,000

The correct answer is b.

Prevalence of a disease is the number of people affected by a condition divided by the total population. In this case, the prevalence is 4,000/1,000,000. Incidence is the number of new cases divided by the population at risk (so it does not include people already affected). In this case, it is 300/996,000. The disease-specific mortality is the number of deaths that can be attributed to the disease divided by the total population. In this case, it is 40/1,000,000.

75. Which of the following represents the incidence of multiple sclerosis:

a) 300/1,000,000
b) 4,000/1,000,000
c) 40/1,000,000
d) 300/996,000
e) 4,000/996,000

The correct answer is d.

Incidence is the number of new cases divided by the population at risk (so it does not include people already affected). In this case, it is 300/996,000. Prevalence of a disease is the number of people affected by a condition divided by the total population. In this case, the prevalence is 4,000/1,000,000. The disease-specific mortality is the number of deaths that can be attributed to the disease divided by the total population. In this case, it is 40/1,000,000.

76. Which of the following represents the disease-specific mortality of multiple sclerosis:

a) 300/1,000,000
b) 4,000/1,000,000
c) 40/1,000,000
d) 300/996,000
e) 4,000/996,000

The correct answer is c.

In this case, the prevalence is 4,000/1,000,000. The disease-specific mortality is the number of deaths that can be attributed to the disease divided by the total population. In this case, it is 40/1,000,000. Incidence is the number of new cases divided by the population at risk (so it does not include people already affected). In this case, it is 300/996,000. Prevalence of a disease is the number of people affected by a condition divided by the total population.

77. Researchers are doing a study to assess the relationship between age and type II diabetes mellitus. They divided the population into three different age groups: under 40, between 40 and 60 and over 60 years old. The percentages of people in each age group who have elevated fasting blood sugar levels will be compared. Which of the following statistical tests would be best suited to this type of study:

a) ANOVA
b) Fisher's exact test

c) Meta-analysis
d) t-test
e) Chi-square test

The correct answer is e.

Chi-square test is used to compare the differences between 2 or more percentages or proportions of categorical outcomes. It is not used to compare mean values. t-test and ANOVA test check the differences between means of groups. Fisher's exact test checks differences between 2 percentages or proportions of categorical outcomes. Meta-analysis is a type of study where researchers pool data from multiple studies for a more precise estimate of the true effect.

78. A group of researchers is trying to determine the prevalence of a disease in the population. Which of the following study types best matches their goal:

a) clinical trial
b) cohort study
c) meta-analysis
d) cross-sectional
e) case-control

The correct answer is d.

Cross-sectional studies are done to determine the prevalence of a disease in the population. As such, they do not establish causality. Case-control studies are done to calculate the odds ratio of exposure to a risk factor between people affected by a condition and those who are unaffected. Cohort studies are done to determine the relative risk of developing a condition depending on the exposure to a risk factor. Clinical trials involve active interventions and are done to compare the tested treatment to either the placebo or the current standard of care. Meta-analysis is a type of study done by pooling the data from previously done studies in order to improve the power and generalizability of the findings.

79. A clinical study is done to assess the efficacy of a new diagnostic test for lung adenocarcinoma. The results are then compared to the results of a biopsy, the gold standard diagnostic test. The number of true positives was 150, the number of false positives was 20, the number of true negatives 130 and the number of false negatives 50. A 65-year-old patient with a 5 week history of dry cough undergoes the test and it comes back positive. Which of the following represents the probability that the patient has lung adenocarcinoma:

a) 150/170
b) 150/200
c) 130/150
d) 130/180
e) 200/350

The correct answer is a.

The positive predictive value represents the likelihood that the person who tests positive for a disease is actually affected. It is calculated as: TP/(TP+FP). The negative predictive value represents the likelihood that the person who tests negative is unaffected. It is calculated as: TN/(TN+FN). Sensitivity is the proportion of people affected who test positive and is calculated as: TP/(TP+FN). Specificity is the proportion of unaffected people who test negative and is calculated as: TN/(TN+FP). Prevalence is calculated as (TP+FN)/(TP+FN+FP+TN). It is not the true prevalence of the disease in the population (as it would require random selection of subjects from the population - which would be impractical for the purposes of the study as the total prevalence of lung adenocarcinoma is low and the number of participants would have to be much higher).

80. Which of the following represents the likelihood that a person who tests negative does not have lung adenocarcinoma:

a) 150/170
b) 150/200
c) 130/150
d) 130/180
e) 200/350

The correct answer is d.

The negative predictive value represents the likelihood that the person who tests negative is unaffected. The positive predictive value represents the likelihood that the person who tests positive for a disease is actually affected. Specificity is the proportion of unaffected people who test negative and is calculated as: TN/(TN+FP). Sensitivity is the proportion of people affected who test positive and is calculated as: TP/(TP+FN). Prevalence is calculated as (TP+FN)/(TP+FN+FP+TN). It is not the true prevalence of the disease in the population (as it would require random selection of subjects from the population - which would be impractical for the purposes of the study as the total prevalence of lung adenocarcinoma is low and the number of participants would have to be much higher).

81. Which of the following represents the probability of an unaffected person testing negative:

a) 150/170
b) 150/200
c) 130/150
d) 130/180
e) 200/350

The correct answer is c.

Specificity is the proportion of unaffected people who test negative and is calculated as: TN/(TN+FP). Sensitivity is the proportion of people affected who test positive and is calculated as: TP/(TP+FN). The positive predictive value represents the likelihood that the person who tests positive for a disease is actually affected. It is calculated as: TP/(TP+FP). The negative predictive value represents the likelihood that the person who tests negative is unaffected. It is calculated as: TN/(TN+FN). Prevalence is calculated as (TP+FN)/(TP+FN+FP+TN). It is not the true prevalence of the disease in the population (as it would require random selection of subjects from the population - which would be impractical for the purposes of the study as the total prevalence of lung adenocarcinoma is low and the number of participants would have to be much higher).

82. Which of the following represents the probability of an affected person testing positive:

a) 150/170
b) 150/200
c) 130/150
d) 130/180
e) 200/350

The correct answer is b.

Sensitivity is the proportion of people affected who test positive and is calculated as: TP/(TP+FN). Specificity is the proportion of unaffected people who test negative and is calculated as: TN/(TN+FP). The positive predictive value represents the likelihood that the person who tests positive for a disease is actually affected. It is calculated as: TP/(TP+FP). The negative predictive value represents the likelihood that the person who tests negative is unaffected. It is calculated as: TN/(TN+FN). Prevalence is calculated as

(TP+FN)/(TP+FN+FP+TN). It is not the true prevalence of the disease in the population (as it would require random selection of subjects from the population - which would be impractical for the purposes of the study as the total prevalence of lung adenocarcinoma is low and the number of participants would have to be much higher).

83. A screening mammography is performed on 2000 women in women older than 50 years. 50 of them are found to have breast cancer. Which of the following best describes the annual incidence and prevalence of breast cancer in this population:

a) incidence cannot be determined, prevalence 2.5%
b) incidence 2.5%, prevalence cannot be determined
c) incidence 2.5%, prevalence 2.5%
d) neither incidence nor prevalence can be determined
e) incidence 0.25%, prevalence 2.5%

The correct answer is a.

Incidence refers to the ratio of the number of newly diagnosed cases in a year and the total number of people at risk. Prevalence is the ratio of the number of people affected by the condition and the total number of people in the population. Prevalence of the disease is therefore 2.5% (50 out of 2000). Incidence cannot be determined from the data given.

84. A medical student is interested in the perception of pain experienced by patients suffering from osteoarthritis. She surveyed 100 patients affected by osteoarthritis. There were 10 questions in the questionnaire, each asking participants to grade their pain from 1-10 at a certain time of the day. The final score was a sum of the 10 pain scores that were reported. 98 of the 100 patients ended the study with final scores between 20 and 50. 2 of the patients had the score of 100. Which of the measures of central tendency is the most affected by these two patients:

a) median
b) mean
c) mode
d) they are all affected equally
e) none of them are affected

The correct answer is b.

Extremely high or low values affect the mean the most as the mean is calculated as the sum of all the values divided by the number of measurements. Median is affected less as it represents the value that has an equal number of measurements that fall below and above it. Mode is affected the least as it represents the most frequently reported value.

85. Which of the measures of central tendency is the least affected by these two patients:

a) median
b) mean
c) mode
d) they are all affected equally
e) none of them are affected

The correct answer is c.

Mode is affected the least as it represents the most frequently reported value. Only two patients reported the values of 100, everyone else fell between 20 and 50. With 98 values falling between 20 and 50, 100 cannot become the mode and the mode is, therefore, unchanged. Extremely high or low values affect the mean the most as the mean is calculated as the sum of all the values divided by the number of measurements. Median is affected less as it represents the value that has an equal number of measurements that fall below and above it.

86. A new biomarker for severe chronic obstructive pulmonary disease (COPD) has been developed. The lower the value of the marker, the higher the probability the patient is affected by COPD. Upon further investigation, it is discovered that it follows a normal distribution. Approximately 2.5% of the target population is affected by severe COPD. Which of the following represents the correct cut-off value for severe COPD in this circumstance:

a) 1 SD below the mean
b) 1 SD above the mean
c) 2 SD above the mean
d) 2 SD below the mean
e) 3 SD below the mean

The correct answer is d.

68% of measurements fall within 1 SD of the mean when the distribution of values is normal. That means that approximately 16% of measurements fall both above and below 1 SD of the mean. 95% of measurements fall within 2 SD's of the mean. Therefore, 2.5% fall under 2 SD's below the mean. 99.7% of measurements fall within 3 SD's of the mean.

87. An epidemiology graduate student is following the recent trends in disease prevalence and incidence. He is noticing that in recent years there has been a significant increase in prevalence of cystic fibrosis. Which of the following best explains the phenomenon:

a) increased incidence due to increased gene prevalence in the population
b) decreased incidence due to infertility of gene carriers
c) increased mortality due to resistant infections
d) lead-time bias
e) improved survival due to better management options

The correct answer is e.

With improved management the prevalence of the disease increases while the incidence remains relatively stable. The management of cystic fibrosis is improving significantly and most patients nowadays reach adulthood and more and more are reaching the age of 50. Cystic fibrosis is usually diagnosed shortly after birth so lead-time bias is not an important factor.

88. Researchers are designing a study to assess whether individuals diagnosed with multiple sclerosis were more likely to have a history of cigarette smoke exposure. Which of the following types of studies best fits that objective:

a) retrospective cohort study
b) prospective cohort study
c) case-control study
d) cross-sectional study
e) clinical trial

The correct answer is c.

Case-control studies are best suited to determine the odds of exposure to a risk factor in affected and unaffected individuals. Cohort studies are best suited to determining the risk of disease development in exposed individuals compared to the unexposed. Cross-sectional studies are done to determine prevalence of the disease. Clinical trials involve an active intervention.

89. A researcher is doing a meta-analysis to assess the relationship between alcohol use and liver cirrhosis. At the end of the study period his results vary significantly from those widely accepted. Which of the following is a possible explanation of the discrepancy:

a) the relationship between alcohol use and liver cirrhosis has changed recently
b) the previously accepted theories could be wrong
c) there may be more than one association between the variables
d) the researcher used low-quality studies for his meta-analysis
e) the power of meta-analysis is often low

The correct answer is d.

The quality of the meta-analysis depends on the quality of the individual studies being used. If the studies that are chosen are biased or have other systemic problems, the findings in the meta-analysis will also be affected. The relationship between alcohol and liver disease does not change over time. Considering how deeply the relationship has been studied in the past, it is also extremely unlikely that all the previous findings were wrong. One of the main advantages of meta-analyses over any other types of clinical research is that their power tends to be much higher as they pool data (and subjects) from a number of different studies.

90. A group of researchers is evaluating the usefulness of plain chest X-Rays in diagnosis lung cancer. They evaluate any shadows based on pre-set criteria. Shadows larger than 2 cm in diameter are referred for further scanning. Which of the following best describes what the effect on findings would be if the cut-off value increased to 3 cm:

a) the number of false positives would not change
b) the number of false positives would increase
c) the number of false negatives would not change
d) the number of false negatives would increase
e) the precision of the study would increase

The correct answer is d.

Larger lesions are more likely to be malignant. Increasing the cut-off value to 3 cm, would likely result in a higher number of false negatives as any cancers that are between 2 and 3 cm in size would be missed. The number of false positives would likely decrease as fewer smaller (and therefore more likely to be benign) lesions would be referred for further testing. Precision refers to how close test results are to one another.

91. Which of the following best describes what effect such change would have on sensitivity and specificity of the test:

a) specificity would increase
b) sensitivity would increase
c) specificity would not change
d) sensitivity would not change
e) incidence would decrease

The correct answer is a.

Increasing the cut-off value means that for those who test positive, the likelihood of the lesion being malignant would be higher. Sensitivity would decrease as a number of affected patients with lesions between 2 and 3 cm in size would be missed. The way a test result is interpreted has no bearing on the incidence of the disease. A way to decrease incidence would be to decrease public exposure to risk factors (eg, smoking cessation programs).

92. A group of researchers is looking into the incidence of ulcerative colitis by age. Which of the following best describes the distribution curve they would most likely see:

a) normal distribution
b) bimodal distribution
c) trimodal distribution
d) positively skewed distribution
e) negatively skewed distribution

The correct answer is b.

Ulcerative colitis has two distinctive peak as far as age distribution goes. It tends to occur either in young adults or in the elderly. Relatively few people develop ulcerative colitis in the middle age. Positively skewed distributions are the ones where mean is larger than the median, which is larger than the mode. Negatively skewed distributions are the opposite. Trimodal distributions would have three peaks. Normal distribution has a single peak with mean, mode and median being the same.

93. A new coronavirus COVID-19 has affected the total of 10,000 people in a population of 100,000. At the end of the epidemic, it was found that on average, 150 new cases were diagnosed every day. The number of people who died from the disease was 100. Which of the following is the case fatality rate of the COVID-19 infection in this population:

a) 10%
b) 1%
c) 1.5%
d) 15%
e) 5%

The correct answer is b.

Case fatality rate is the percentage of people affected who end up dying from the disease. In this case, of the 10,000 people affected, 100 died. That gives the case fatality rate of 1%. Incidence refers to the number of new cases among the number of people at risk, while the prevalence refers to the total number of people affected divided by the total population.

94. A 58-year-old male undergoes colorectal cancer screening with fecal occult blood test. He is relieved to find out that the results of the test are negative, although he worries that they may be false negative as multiple of his family members have suffered from the disease. He is concerned that the relatively low prevalence of the disease in the population affects the results of the test. Which of the following statements is true in regards to that:

a) the low prevalence of the disease increases the sensitivity of the test
b) the low prevalence of the disease decreases the sensitivity of the test
c) the low prevalence of the disease does not affect the sensitivity of the test
d) the low prevalence of the disease does not affect the negative predictive value of the test
e) the low prevalence of the disease decreases the negative predictive value of the test

The correct answer is c.

Sensitivity and specificity are inherent properties of a diagnostic test and they do not depend on the prevalence of the disease in population. Sensitivity of the test is the probability that a person affected by a condition tests positive. In an ideal scenario, a sensitivity of 100% would indicate that everyone affected by a condition would get a positive test result. Positive and negative predictive value of the test, on the other hand, do depend on the prevalence of the disease in question. They indicate what the likelihood is that someone who tests positive or negative, respectively, is or is not affected by the condition, respectively. For diseases with relatively low prevalence, the negative predictive value of a test is high, meaning that people who test negative are very likely not to be have the disease.

95. Which of the following would change if the disease in question affected a significantly larger percentage of the population:

a) the negative predictive value would increase
b) the negative predictive value would not change
c) the sensitivity would increase
d) the positive predictive value would increase
e) the specificity would increase

The correct answer is d.

Increasing the prevalence of the disease would increase the positive predictive value of the test. The positive predictive value refers to the probability that a person who tests positive is affected by the condition. The more prevalent the disease is the more likely everyone is to have it, regardless of the way they test. The effect on the negative predictive value would be the opposite. Sensitivity and specificity would not change as they are inherent properties of the test.

96. In general, which of the following increases the sensitivity of a test:

a) increased prevalence of the disease
b) decreased incidence of the disease
c) increased number of tests performed
d) narrowing the definition of a positive result
e) widening the definition of a positive result

The correct answer is e.

Widening the definition of a positive result for further testing increases the number of people who test positive. Sensitivity of the test refers to the probability that a person affected by a condition would get a positive test result. In an ideal scenario, everyone with a condition would get a positive test result, indicating the sensitivity of 100%. However, in practice, there will be values which are extraordinarily infrequent among people affected by the condition and further testing in that group would result in extremely high false positive rates. It would, however, increase the sensitivity of the test as more people affected by the condition would test positive.

Prevalence, incidence and the number of tests that are performed do not affect the sensitivity of the test.

97. In a survey of 200 households with the average of 3 residents per household, 150 individuals with hypertension are detected. Which of the following is the best estimate of the prevalence of hypertension in this population:

a) 15%
b) 25%
c) 33%
d) 67%
e) 75%

The correct answer is b.

The total of 600 people were surveyed (200 households * 3 residents). 150 of them were found to be hypertensive. The prevalence of the disease is calculated as the number of people affected by a condition divided by the total number of people at risk. 150/600 = 0.25 = 25%.

98. A group of researchers is studying the association between radiation exposure and acute myeloid leukemia (AML) development. They have identified a group of patients with AML and a similar group of patients without AML. History of radiation exposure is obtained from both groups. Which of the following best describes such a study:

a) case-control
b) cross-sectional
c) randomized clinical trial
d) prospective cohort
e) retrospective cohort

The correct answer is a.

Case-control studies assess the odds ratio of exposure to a risk factor between the group exposed to a risk factor and a similar group unexposed to the same risk factor. Controls have to be similar to cases in their baseline characteristics. Cross-sectional studies are done to assess prevalence of a disease. Randomized clinical trials involve an active intervention. Cohort studies are also done by identifying a group exposed to a risk factor and a group unexposed to the same factor. The two groups are then followed for a period of time for the development of an effect. In retrospective cohort studies the start date is in the past.

99. Researchers have developed a new diagnostic test for the COVID-19 infection. The sensitivity of the test is reported to be 65% and the specificity is 95%. The test was performed on 100 patients affected by COVID-19 and 100 patients who are unaffected. Which of the following is the number of false negative test results:

a) 5
b) 35
c) 50
d) 65
e) 95

The correct answer is b.

The sensitivity of the test is calculated as TP/(TP+FN). The specificity is calculated as TN/(TN+FP). The number of people who have the disease is TP+FN. The number of people who do not have the disease is TN+FP. The positive predictive value is calculated as TP/(TP+FP). The negative predictive value is calculated as TN/(TN+FN)
TP/(TP+FN) = 0.65 => TP/100 = 0.65 => TP = 65 => FN = 100 - 65 = 35
TN/(TN+FP) = 0.95 => TN/100 = 0.95 => TN = 95 => FP = 100 - 95 = 5
PPV = TP/(TP+FP) = 65/(65+5) = 65/70 = 93%
NPV = TN/(TN+FN) = 95/(95+35) = 95/130 = 73%

100. Which of the following is the number of false positive test results:

a) 5
b) 35
c) 50
d) 65
e) 95

The correct answer is a.

The sensitivity of the test is calculated as TP/(TP+FN). The specificity is calculated as TN/(TN+FP). The number of people who have the disease is TP+FN. The number of people who do not have the disease is TN+FP. The positive predictive value is calculated as

TP/(TP+FP). The negative predictive value is calculated as TN/(TN+FN)
TP/(TP+FN) = 0.65 => TP/100 = 0.65 => TP = 65 => FN = 100 - 65 = 35
TN/(TN+FP) = 0.95 => TN/100 = 0.95 => TN = 95 => FP = 100 - 95 = 5
PPV = TP/(TP+FP) = 65/(65+5) = 65/70 = 93%
NPV = TN/(TN+FN) = 95/(95+35) = 95/130 = 73%

101. Which of the following is the positive predictive value of the test:

a) 35%
b) 65%
c) 73%
d) 93%
e) 95%

The correct answer is d.

The sensitivity of the test is calculated as TP/(TP+FN). The specificity is calculated as TN/(TN+FP). The number of people who have the disease is TP+FN. The number of people who do not have the disease is TN+FP. The positive predictive value is calculated as TP/(TP+FP). The negative predictive value is calculated as TN/(TN+FN)
TP/(TP+FN) = 0.65 => TP/100 = 0.65 => TP = 65 => FN = 100 - 65 = 35
TN/(TN+FP) = 0.95 => TN/100 = 0.95 => TN = 95 => FP = 100 - 95 = 5
PPV = TP/(TP+FP) = 65/(65+5) = 65/70 = 93%
NPV = TN/(TN+FN) = 95/(95+35) = 95/130 = 73%

102. Which of the following is the negative predictive value of the test:

a) 35%
b) 65%
c) 73%
d) 93%
e) 95%

The correct answer is c.

The sensitivity of the test is calculated as TP/(TP+FN). The specificity is calculated as TN/(TN+FP). The number of people who have the disease is TP+FN. The number of people who do not have the disease is TN+FP. The positive predictive value is calculated as

TP/(TP+FP). The negative predictive value is calculated as TN/(TN+FN)
TP/(TP+FN) = 0.65 => TP/100 = 0.65 => TP = 65 => FN = 100 - 65 = 35
TN/(TN+FP) = 0.95 => TN/100 = 0.95 => TN = 95 => FP = 100 - 95 = 5
PPV = TP/(TP+FP) = 65/(65+5) = 65/70 = 93%
NPV = TN/(TN+FN) = 95/(95+35) = 95/130 = 73%

103. Which of the following would increase the positive predictive value of the test:

a) better treatment of the disease
b) improved patient isolation
c) patients refusing to self-isolate
d) vaccine development
e) increased number of ICU beds

The correct answer is c.

Increased prevalence of the disease leads to increased positive predictive value of the tests because people in general are more likely to be affected by the condition regardless of the way they test. None of the other answer choices would significantly increase the prevalence of the disease.

104. A retrospective cohort trial was done to assess the relationship between long-term aspirin use and colorectal carcinoma development. The reported relative risk was 0.82 with the 95% confidence interval 0.76 to 0.96. Which of the following ‘p’ values best correlates with those findings:

a) 0.03
b) 0.07
c) 0.1
d) 0.5
e) 1.0

The correct answer is a.

The relative risk determines the risk of disease development in the exposed group versus that in the unexposed group. Values lower than 1 indicate that the factor being studied is protective. If the value is 1, it indicates that there is no statistically significant association between the variables being studied. The 95% confidence must not contain the value 1 in order for results to be statistically significant. When the confidence interval does not contain the value 1, it corresponds to the p value less than 0.05.

105. The same trial was done simultaneously at a different center. They found the relative risk to be 0.87 with the 95% confidence interval 0.70 to 1.15. They also reported that they lost a significant percentage of the population to follow-up, mostly among those who were not taking aspirin. Which of the following biases is most likely affecting their results:

a) recall bias
b) observer bias
c) lead-time bias
d) selection bias
e) length-time bias

The correct answer is d.

Attrition bias is a form of selection bias. It occurs when a large number of participants are lost to follow up and the majority of them are from the same group, altering the findings in the study. In this study, it is possible that the uneven attrition of participants led to the loss of statistical significance of the findings. Recall bias occurs due to tendency of people to overreport exposures once the disease develops. Observer bias occurs when the researcher's strong belief in the existence or non-existence of an association affects the way the data is analyzed. Lead-time bias refers to the appearance of improved survival due to earlier detection of disease without actually altering its course. Length-time bias refers to the tendency of screening tests to diagnose diseases with longer latency periods.

106. The investigators are looking into the average fasting blood glucose levels among patients in a primary care clinic. The clinic takes care of 1000 patients and it was found that fasting blood glucose levels were normally distributed with a mean of 90 mg/dL and standard deviation of 10 mg/dL. Which of the following represents the number of patients who have blood glucose levels >110 mg/dL:

a) 2
b) 10
c) 25
d) 50
e) 160

The correct answer is c.

With the mean of 90 and SD of 10, the values above 110 mg/dL represent > 2 SDs above the mean. Approximately 95% of patients have values within 2 SDs of the mean. That means that approximately 2.5% of patients will have values that are either above or below 2 SDs of the mean. 2.5% of 1000 participants is 25 participants. Roughly 68% of the population will have values within 1 SD of the mean, meaning that 16% (or 160 patients) will have values above 1 SD of the mean (100 mg/dL). Between 1 and 2 patients will have values above 120 mg/dL (3 SDs above the mean).

107. A large clinical trial was done to assess the efficacy of the new diabetes drug. In the first phase of the trial, several volunteers were given the drug and no serious side effects were seen. Which of the following was likely assessed during this phase of the trial:

a) efficacy
b) dosing
c) comparison with standard of care
d) pharmacokinetics
e) black box warnings

The correct answer is d.

Safety, toxicity, pharmacokinetics and pharmacodynamics are assessed in the first phase of clinical trials, when the drug is typically given to a smaller number of volunteers. In the second phase, the drug is given to a larger number of patients and the efficacy, dosing, and adverse effects are assessed. After that, in phase III it is compared to placebo or the current standard of care. Finally, once the drug is released to the market it is still followed for years to assess for rare or long-term adverse effects (eg, black box warnings).

108. Which of the following would most likely be assessed in the next phase of the trial if the drug successfully passes this phase:

a) comparison with standard of care
b) dosing
c) pharmacokinetics
d) pharmacodynamics
e) black box warnings

The correct answer is b.

In the second phase, the drug is given to a larger number of patients and the efficacy, dosing, and adverse effects are assessed. After that, in phase III it is compared to placebo or the current standard of care. Finally, once the drug is released to the market it is still followed for years to assess for rare or long-term adverse effects (eg, black box warnings). Pharmacokinetics and pharmacodynamics are assessed in the first phase of the clinical trial.

109. A group of public health officials is looking into developing the new screening program for aggressive prostate cancer. Which of the following features of the test would make it most appropriate as the screening test of choice:

a) sensitivity 65%, specificity 95%
b) sensitivity 85%, specificity 85%
c) sensitivity 75%, specificity 35%
d) sensitivity 95%, specificity 30%
e) sensitivity 35%, specificity 90%

The correct answer is d.

The purpose of screening is to identify as many people as possible who have the disease, even at the expense of getting a larger number of false positive results. The number of false negatives needs to remain low. Those are the features of a highly sensitive test. In this case, test d is the most sensitive and, therefore, the best for screening. Following a positive screening test, a confirmatory, highly specific test can be done to definitively diagnose the condition. In this case, the most specific test is test a.

110. Which of the tests would be the most appropriate confirmatory test:

a) sensitivity 65%, specificity 95%
b) sensitivity 85%, specificity 85%
c) sensitivity 75%, specificity 35%
d) sensitivity 95%, specificity 30%
e) sensitivity 35%, specificity 90%

The correct answer is a.

Confirmatory tests need to be highly specific, meaning that they correctly identify people who do not have the condition in question. The number of false positives needs to remain low (vs number of false negatives that needs to remain low in screening tests). In this case, the most specific test is test a.

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